Obesity Surgery (2025) 35:4628–4638
https://doi.org/10.1007/s11695-025-08249-x

RESEARCH

Evaluating Artificial Intelligence-Generated Patient Education Materials for Bariatric Surgery: Comparative Analysis of Response Quality, Reliability, and Readability Across ChatGPT and DeepSeek Models

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Received: 19 June 2025 / Revised: 22 August 2025 / Accepted: 30 August 2025 / Published online: 27 September 2025

Abstract

Background Artificial intelligence (AI) models such as ChatGPT and DeepSeek have gained increasing attention for their potential to enhance patient education by delivering accessible and evidence-based health information. We designed the following study to evaluate the AI models—ChatGPT and DeepSeek—in generating patient education materials for bariatric surgery.

Methods Thirty commonly asked patient questions related to bariatric surgery were classified into four thematic domains: (1) surgical planning and technical considerations, (2) preoperative assessment and optimization, (3) postoperative care and complication management, and (4) long-term follow-up and disease management. Responses generated by ChatGPT and DeepSeek were evaluated using three key metrics: (1) response quality, assessed by the Global Quality Score, rated on a 5-point scale from 1 (poor) to 5 (excellent); (2) reliability, measured using modified DISCERN criteria, which assess adherence to clinical guidelines and evidence-based standards, with scores ranging from 5 (low) to 25 (high); and (3) readability, evaluated using two validated formulas: the Flesch-Kincaid Grade Level and the Flesch Reading Ease Score.

Results ChatGPT significantly outperformed DeepSeek in response quality, with a median (IQR) Global Quality Score of 5.00 (4.00, 5.00) vs. 4.00 (4.00, 5.00) ($P=0.002$). Higher reliability was also observed in ChatGPT, as reflected by mDISCERN scores across all four domains (median [IQR], 22.0 [21.0, 23.25] vs. 19.7 [19.0, 20.75]; $P<0.001$). While no significant difference was found in the Flesch Reading Ease Score (mean [SD], 26.11 [12.84] vs. 20.87 [12.20]; $P=0.110$), ChatGPT yielded significantly higher Flesch-Kincaid Grade Level Scores (meaning its text was more complex) (mean [SD], 16.40 [2.43] vs. 13.48 [2.35]; $P<0.001$). Both models produced responses at a readability level corresponding to college education.

Conclusions ChatGPT provided higher-quality and more reliable responses, while DeepSeek's answers were slightly easier to read. However, both models' answers lacked attention to psychosocial and cultural aspects of patient care, highlighting the need for more empathetic, adaptive AI to support inclusive patient education.

Keywords AI · Bariatric surgery · Patient education · ChatGPT · DeepSeek

Introduction

Over the past two decades, the worldwide prevalence of obesity—and particularly severe obesity—has been steadily increasing, representing a growing public health concern [1]. Bariatric (metabolic) surgery has become a well-established and effective treatment for severe obesity, leading to significant and sustained weight loss, improvements in metabolic parameters, and reduced incidence of obesity-related comorbidities such as type 2 diabetes, hypertension, and obstructive sleep apnea [2]. The global rise in obesity and the corresponding increase in bariatric surgeries emphasize the need for advanced strategies to enhance patient care and healthcare system efficiency. Due to the complexity of bariatric procedures and their long-term effects, patients require clear guidance and sustained support before, during, and

Extended author information available on the last page of the article after surgery. This underscores the importance of effective communication, education, and accessible resources to improve patient empowerment and clinical outcomes [3].

The application of AI in medicine has expanded rapidly in recent years, driven by the development of diverse technologies aimed at enhancing patient education and promoting evidence-based clinical practice [4, 5]. Among these, large language models (LLMs), such as ChatGPT and DeepSeek, have demonstrated considerable potential to transform patient communication by utilizing extensive datasets to generate accurate, conversational, and easily comprehensible responses to user queries [6, 7]. ChatGPT is one of the most advanced LLMs developed by OpenAI, designed to generate text-based outputs by integrating information from sources such as scientific articles, books, journals, and open-access websites. It is also capable of performing a wide range of language-related tasks, including translation and responding to user queries in a contextually appropriate manner [8]. DeepSeek, a China-based open-source LLM that emphasizes transparency and reproducibility, gained global attention following its December 2024 launch due to its low cost and high download volume [9]. Despite their promising capabilities, these applications have notable limitations, including potential inaccuracies due to reliance on web-sourced content and insufficient depth in certain responses [10].

Given the rapid advancements and inherent limitations of AI-driven language models, this study aimed to compare the performance of ChatGPT and DeepSeek in the context of patient education related to bariatric surgery. The primary focus is on evaluating the overall response quality, objectivity, comprehensiveness of medical information, and readability. Furthermore, the study examines the models' ability to address a broad range of patient concerns, categorized into four key domains: (1) surgical planning and technical considerations, (2) preoperative assessment and optimization, (3) postoperative care and complication management, and (4) long-term follow-up and disease management. Through this comparative analysis, we aim to explore the potential and limitations of LLMs in improving patient-centered education in bariatric surgery.

Methods

Study Design and Question Selection

This comparative study evaluated the performance of ChatGPT and DeepSeek in responding to 30 patient-centered questions related to bariatric surgery. The questions were sourced from the "Frequently Asked Questions" section of the American Society for Metabolic and Bariatric Surgery (ASMBS), ensuring clinical relevance and alignment with common patient concerns.

To enhance representativeness, the questions were designed to reflect the informational needs of diverse patient populations, taking into account variations in educational background, cultural context, and gender-specific issues. Supplementary eTable 1 provides the complete list of questions and model responses.

Each question was systematically selected through a comprehensive review of existing literature and organized into four thematic domains [11]:

1) Surgical planning and technical considerations (Q1, Q2, Q3, Q6, Q10, Q17, Q20)
2) Preoperative assessment and optimization (Q8, Q11, Q26, Q27, Q28, Q29, Q30)
3) Postoperative care and complication management (Q5, Q7, Q9, Q12, Q13, Q14, Q16, Q18)
4) Long-term follow-up and disease management (Q4, Q15, Q19, Q21, Q22, Q23, Q24, Q25)

Each of the 30 questions was independently entered into both ChatGPT and DeepSeek using the "New Chat" function to avoid contextual memory effects. The version of ChatGPT used was GPT-4 (April 2025), while the DeepSeek responses were generated using R1 (January 2025).

The responses from both models were evaluated by three independent raters in gastrointestinal surgery, all of whom are board-certified bariatric surgeons with substantial clinical and academic experience. To minimize potential bias, all responses were anonymized and coded prior to evaluation, and the source model (ChatGPT or DeepSeek) was not disclosed. Each rater received the question–answer sets in a different randomized order and independently assessed each response based on predefined criteria.

Evaluation Metrics

Response Quality

The overall quality of the responses was assessed using the Global Quality Score (GQS), a validated five-point scale in which higher scores indicate better quality. Scores range from 1 (poor) to 5 (excellent) (Table 1) [12].

Reliability

The DISCERN tool, originally developed by Charnock et al., is a validated tool commonly used to assess the quality of consumer health information [13]. In this study, a modified version of DISCERN (mDISCERN) was used to evaluate the objectivity, comprehensiveness, and evidence-based nature of the responses. The mDISCERN consists of five items, each addressing a key aspect of medical information quality and scored from 1 to 5, with a total

Table 1 Global Quality Scale (GQS) criteria

Score	Description
1	Poor quality and flow, most information missing, not at all useful for patients
2	Generally poor quality and flow, some information listed but many important topics missing, of very limited use to patients
3	Moderate quality, suboptimal flow, some important information is adequately discussed but others poorly discussed, somewhat useful for patients
4	Good quality and generally good flow, most of the relevant information is listed, but some topics are not covered, useful for patients
5	Excellent quality and excellent flow, very useful for patients

Table 2 Modified DISCERN (mDISCERN) criteria

Item	Description	Score
1	Are the aims clear and achieved?	1–5 points
2	Are reliable sources of information used? (i.e., publication cited)	1–5 points
3	Is the information presented balanced and unbiased?	1–5 points
4	Are additional sources of information listed for patient reference?	1–5 points
5	Are areas of uncertainty mentioned?	1–5 points
Total		5–25 points

Table 3 Flesch-Kincaid readability scores

Score	Grade level	Summary
90–100	5th grade	Very easy to read
80–90	6th grade	Easy to read
70–80	7th grade	Fairly easy to read
60–70	8th & 9th grade	Plain English
50–60	10th to 12th grade	Fairly difficult to read
30–50	College	Difficult to read
10–30	College graduate	Very difficult to read
0–10	Professional	Extremely difficult to read

score between 5 and 25. Higher scores reflect greater reliability, objectivity, and adherence to established clinical guidelines (Table 2).

Readability

Readability of the responses was evaluated using two validated formulas: the Flesch-Kincaid Grade Level (FKGL) and the Flesch Reading Ease Score (FRES). FKGL estimates the US school grade level required to comprehend the text, while the FRES ranges from 0 to 100, with higher scores indicating easier readability.

These formulas were selected due to their widespread use and established validity in health communication research. In accordance with recommendations from the American Medical Association (AMA), the target readability level was set between the sixth and eighth grades to ensure accessibility for the general patient population (Table 3) [14].

Statistical Analysis

Statistical analyses were performed using SPSS version 26.0 (IBM Corp., Armonk, NY, USA). The normality of continuous variables was assessed using the Shapiro–Wilk test. Variables with normal distribution were presented as mean and standard deviation (SD), and comparisons between groups were conducted using the independent samples *t*-test. For variables not normally distributed, data were expressed as median and interquartile range (IQR), and comparisons were made using the Mann–Whitney *U* test. Inter-rater reliability among the three raters was assessed using intraclass correlation coefficients (ICC). The ICC values were interpreted according to established guidelines: excellent (>0.90), good (0.75–0.90), moderate (0.50–0.75), and poor (<0.50) [15]. When the ICC was ≥ 0.75, indicating good or excellent inter-rater agreement, the mean score per question across raters was calculated as an overall score. Group comparisons were then conducted based on the distribution of these overall scores: if normally distributed, independent samples *t*-tests were used; otherwise, the Mann–Whitney *U* test was applied. In contrast, when the ICC was <0.75, indicating only moderate or poor agreement, linear mixed-effects models were applied to account for variability between raters when comparing the two groups. A P-value <0.05 was considered statistically significant.

Results

Response Quality

Across all three raters, ChatGPT consistently outperformed DeepSeek. The median GQS for ChatGPT was 5.00 (IQR 4.00, 5.00), while DeepSeek had a median of 4.00 (IQR 4.00, 5.00). Statistical comparisons showed significant differences between the two models for all individual raters (Rater 1 $P = 0.019$; Rater 2 $P = 0.037$; Rater 3 $P = 0.015$) and the overall score (Difference − 0.36, 95% CI − 0.58 to − 0.14, $P = 0.002$), favoring ChatGPT (Table 4, Fig. 1). In the comparison of four thematic domains, ChatGPT showed significantly higher GQS in surgical planning and technical consideration (Difference − 0.52, 95% CI − 0.89 to − 0.16, $P = 0.006$) and postoperative care and complications (Difference − 0.46,

Table 4 Comparison of ChatGPT and DeepSeek on response quality, reliability, and readability

Metric		DeepSeek	ChatGPT	Difference (95% CI)	*P value*
GQS (median, IQR)	Rater 1	4.00 (4.00, 5.00)	5.00 (4.00, 5.00)	− 0.37 (− 0.66 to − 0.08)	0.019
	Rater 2	4.00 (4.00, 5.00)	5.00 (4.00, 5.00)	− 0.37 (− 0.69 to − 0.01)	0.037
	Rater 3	4.00 (4.00, 5.00)	5.00 (4.00, 5.00)	− 0.33 (− 0.58 to − 0.07)	0.015
	Overall[a]	4.00 (4.00, 5.00)	5.00 (4.00, 5.00)	− 0.36 (− 0.58 to − 0.14)	0.002
mDISCERN (median, IQR)	Rater 1	20.00 (19.00, 21.00)	23.00 (22.00, 24.00)	− 2.87 (− 3.63 to − 2.01)	< 0.001
	Rater 2	20.00 (18.00, 21.00)	23.00 (21.00, 24.00)	− 2.70 (− 3.66 to − 1.76)	< 0.001
	Rater 3	20.00 (18.75, 21.00)	22.00 (21.00, 23.25)	− 2.63 (− 3.53 to − 1.77)	< 0.001
	Overall[b]	19.67 (19.00, 20.75)	22.50 (21.67, 24.00)	− 2.73 (− 3.56 to − 1.88)	< 0.001
Reading Score (mean, SD)		26.11 (12.84)	20.87 (12.20)	5.24 (− 1.23 to 11.72)	0.110
Grade Level Score (mean, SD)		13.48 (2.35)	16.40 (2.43)	− 2.92 (− 4.15 to − 1.68)	< 0.001

[a]Given that the intraclass correlation coefficients (ICC) for both groups (DeepSeek = 0.47; ChatGPT = 0.57) were below 0.75, indicating poor or moderate inter-rater reliability, linear mixed-effects models were used to assess the differences between two groups

[b]The ICC for both groups (DeepSeek = 0.94; ChatGPT = 0.89) were above 0.75, demonstrating excellent inter-rater reliability. Therefore, the mean score across raters was calculated for each question, and group differences were assessed using the Mann–Whitney *U* test due to non-normal distribution

Table 5 Comparison of ChatGPT and DeepSeek on by question type

Metric		Surgical planning and technical consideration	Preoperative assessment and optimization	Postoperative care and complications	Long-term follow-up and management
GQS (median, IQR)	DeepSeek	4.00 (4.00, 4.50)	5.00 (4.00, 5.00)	4.00 (4.00, 4.75)	4.00 (4.00, 5.00)
	ChatGPT	5.00 (4.00, 5.00)	5.00 (5.00, 5.00)	5.00 (4.00, 5.00)	5.00 (4.00, 5.00)
	Difference (95% CI)	− 0.52 (− 0.89 to − 0.16)	− 0.24 (− 0.51 to 0.03)	− 0.46 (− 0.80 to − 0.12)	− 0.25 (− 0.59 to 0.09)
	P value	0.006	0.081	0.010	0.150
mDISCERN (median, IQR)	DeepSeek	19.67 (17.33, 21.00)	20.67 (19.33, 21.67)	19.84 (19.67, 20.00)	19.67 (17.58, 21.00)
	ChatGPT	23.67 (21.67, 24.00)	23.00 (22.00, 23.67)	22.50 (22.00, 24.50)	21.67 (20.75, 23.25)
	Difference (95% CI)	− 3.24 (− 5.14 to − 1.05)	− 2.19 (− 3.25 to − 0.89)	− 3.13 (− 4.60 to − 1.68)	− 2.29 (− 4.09 to − 0.37)
	P value	0.015	0.009	0.004	0.027
Reading Score (mean, SD)	DeepSeek	25.32 (7.00)	23.72 (13.00)	27.14 (19.74)	28.91 (10.50)
	ChatGPT	17.16 (10.45)	15.55 (7.96)	19.99 (15.86)	31.94 (7.84)
	Difference (95% CI)	8.16 (− 2.19 to 18.52)	8.17 (− 4.38 to 20.73)	7.15 (−12.05 to 26.35)	− 3.03 (− 12.96 to 6.91)
	P value	0.112	0.181	0.438	0.524
Grade Level Score (mean, SD)	DeepSeek	13.23 (1.01)	14.44 (2.73)	12.88 (3.20)	13.19 (2.15)
	ChatGPT	17.31 (2.50)	17.13 (2.13)	15.96 (1.76)	14.64 (2.67)
	Difference (95% CI)	− 4.08 (− 6.30 to − 1.86)	− 2.69 (− 5.55 to 0.16)	− 3.09 (− 5.86 to − 0.32)	− 1.45 (− 4.05 to 1.16)
	P value	0.002	0.062	0.031	0.254

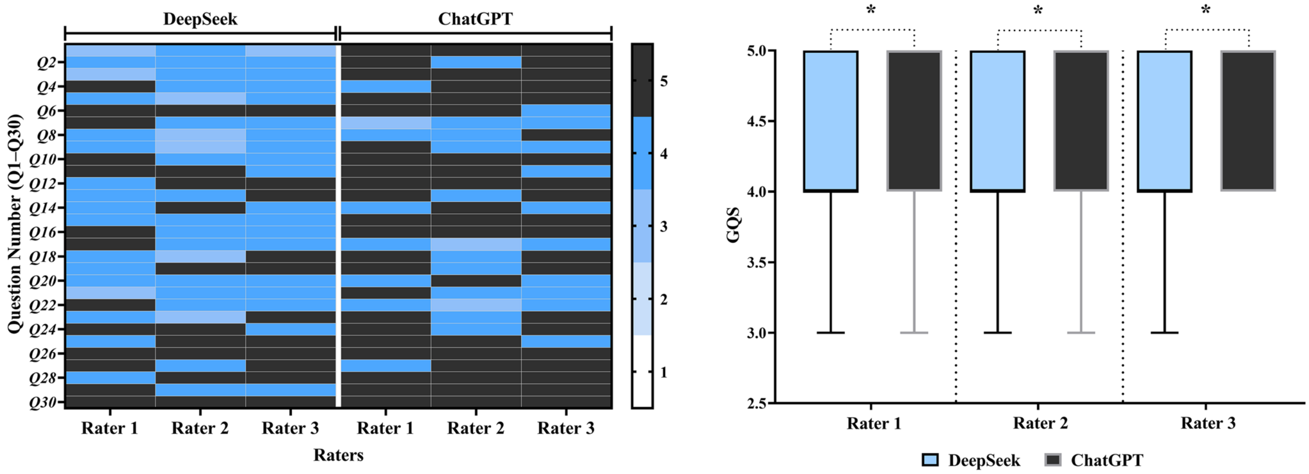

Fig. 1 Comparison of response quality between ChatGPT and DeepSeek. The left panel is a heatmap showing GQS assigned to each of the 30 questions (rows) by three independent raters (columns), for both DeepSeek and ChatGPT. The color gradient reflects the score values from 1 (poor) to 5 (excellent), with darker colors indicating higher quality. The right panel presents boxplots summarizing overall GQS distributions per rater, comparing the two models. Boxplots present the median and IQR; statistical significance is indicated as $* = P < 0.05$ (ChatGPT vs DeepSeek)

95% CI − 0.80 to − 0.12, $P = 0.010$) (Table 5, Fig. 4). ChatGPT's responses were more comprehensive and structured, particularly in addressing technically complex or safety–critical topics.

ChatGPT consistently provided higher-quality responses compared to DeepSeek. Representative examples of model outputs illustrate these differences. For instance, in response to the question *"Is weight loss surgery safe?"*, ChatGPT demonstrated a notably more informative and patient-oriented approach. ChatGPT's answer included quantitative data, such as a 0.1% perioperative mortality rate and a 3–5% incidence of serious complications, thereby enhancing credibility and context. Moreover, ChatGPT identified high-risk populations (e.g., older patients, individuals with multiple comorbidities) and emphasized key factors influencing surgical safety, including surgeon experience and type of procedure. Importantly, it also highlighted the role of multidisciplinary care teams in optimizing perioperative outcomes. In addition to risk discussion, ChatGPT addressed long-term benefits of bariatric surgery, such as reductions in cardiovascular mortality and diabetes remission, offering a balanced and evidence-based perspective suitable for patient education.

In contrast, DeepSeek provided a structurally organized yet more generalized response. While the classification of surgical procedures was clearly presented, the answer lacked quantitative data, references, or discussion of modifiable outcome predictors such as surgeon experience. The response was more descriptive in tone and did not address multidisciplinary support or specific long-term health benefits.

Overall, ChatGPT's response resembled a comprehensive patient education resource, combining objectivity with an encouraging tone. DeepSeek's answer, on the other hand, was more akin to a concise clinical summary, potentially more suitable for healthcare professionals seeking a high-level overview.

Reliability

ChatGPT achieved higher mDISCERN median scores compared to DeepSeek across all raters, with median (IQR) scores of 22 (21, 23.25) vs. 19.7 (19, 20.75). The differences were statistically significant for all raters ($P < 0.001$) as well as the overall comparison (Difference − 2.73, 95% CI − 3.56 to − 1.88, $P < 0.001$) (Table 4, Fig. 2). Meanwhile, ChatGPT also maintained comparatively strong performance across all thematic domains (Table 5, Fig. 4). These results indicate ChatGPT has improved its ability to provide evidence-based, clinically guideline-adherent responses.

Compared to DeepSeek, ChatGPT generated greater reliability and objectivity responses. For example, when asked, *"What's the sleeve gastrectomy?"*, DeepSeek tended to deliver answers with a clear, structured format, often segmented by headings. This approach enhanced information density and readability, resembling an encyclopedia-style entry. The language used leaned toward a more technical, medically descriptive tone, making it suitable for readers with a clinical background.

By contrast, ChatGPT provided equally comprehensive responses but adopted a more conversational and narrative structure, akin to a medical Q&A article. The content was more user-centric, often including comparative information—such as distinctions between sleeve gastrectomy, Roux-en-Y gastric bypass (RYGB), and adjustable gastric

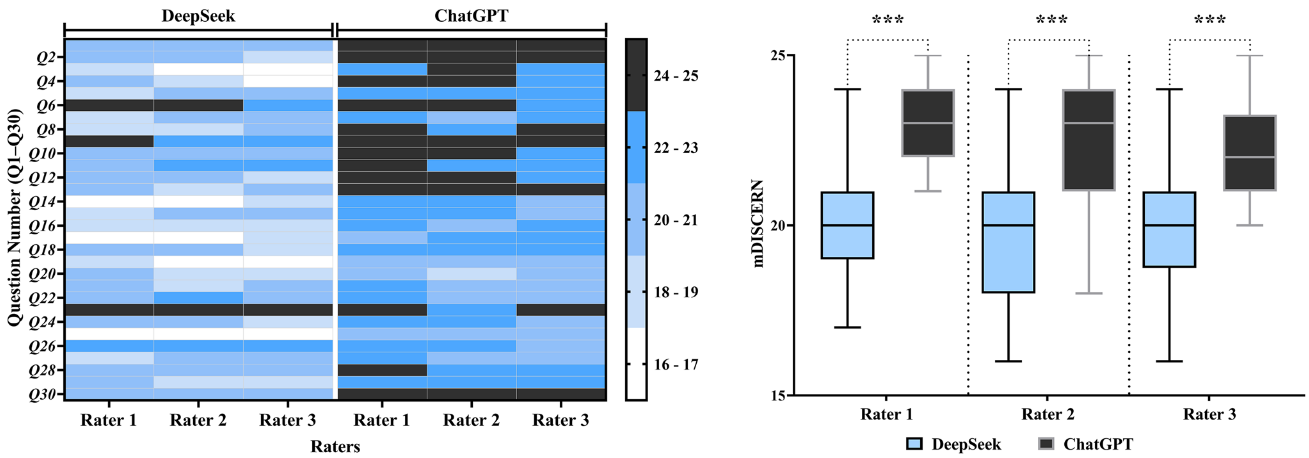

Fig. 2 Comparison of reliability between ChatGPT and DeepSeek. The left panel is a heatmap showing mDISCERN score assigned to each of the 30 questions (rows) by three independent raters (columns), for both DeepSeek and ChatGPT. As all observed scores fell between 16 and 25, the color gradient was divided into five intervals: 16–17, 18–19, 20–21, 22–23, and 24–25, with darker colors indicating higher reliability. The right panel presents boxplots summarizing overall mDISCERN score distributions per rater, comparing the two models. Boxplots present the median and IQR; statistical significance is indicated as *** = $P < 0.001$ (ChatGPT vs DeepSeek)

banding—and contextualized with authoritative references from sources like the Mayo Clinic and the NHS. This reinforced both accessibility and credibility.

In summary, ChatGPT prioritized practical relevance, understandability, and authoritative sourcing, making it particularly effective for patient education. Meanwhile, DeepSeek exhibited a more systematic and encyclopedic style, with strengths in technical summarization.

Readability

No significant difference in FRES was observed between ChatGPT and DeepSeek (difference 5.24; 95% CI − 1.23 to 11.72; $P = 0.110$). Both models Generated responses at a readability level corresponding to college education. The FKGL was 16.40 for ChatGPT and 13.48 for DeepSeek (Table 4, Fig. 3). Although the difference between the two models was statistically significant (− 2.92; 95% CI − 4.15 to − 1.68; $P < 0.001$), both exceeded the recommended sixth- to eighth-grade reading level for patient education materials, as outlined by the AMA. Similar results were also observed across the four thematic domains (Table 5, Fig. 4). While the outputs may be appropriate for a general college audience, the elevated reading levels suggest that further simplification is necessary to enhance accessibility, particularly for patients with limited health literacy.

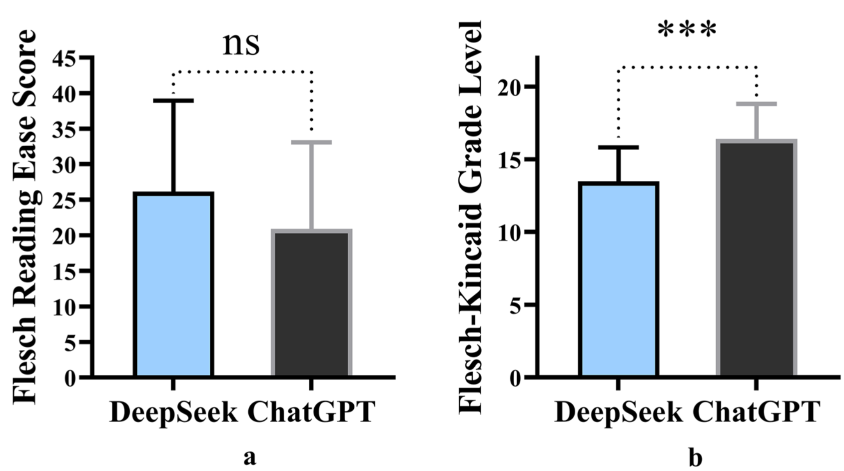

Fig. 3 Comparison of readability between ChatGPT and DeepSeek. **a** The Flesch Reading Ease Score (FRES) comparison between ChatGPT and DeepSeek; higher FRES indicates easier-to-read text. **b** The Flesch-Kincaid Grade Level (FKGL) comparison between ChatGPT and DeepSeek; higher FKGL indicates more complex text requiring a higher reading level. All data are expressed as the mean with SD and statistical significance is indicated as ns = not statistically, *** = $P <$ 0.001 (ChatGPT vs DeepSeek)

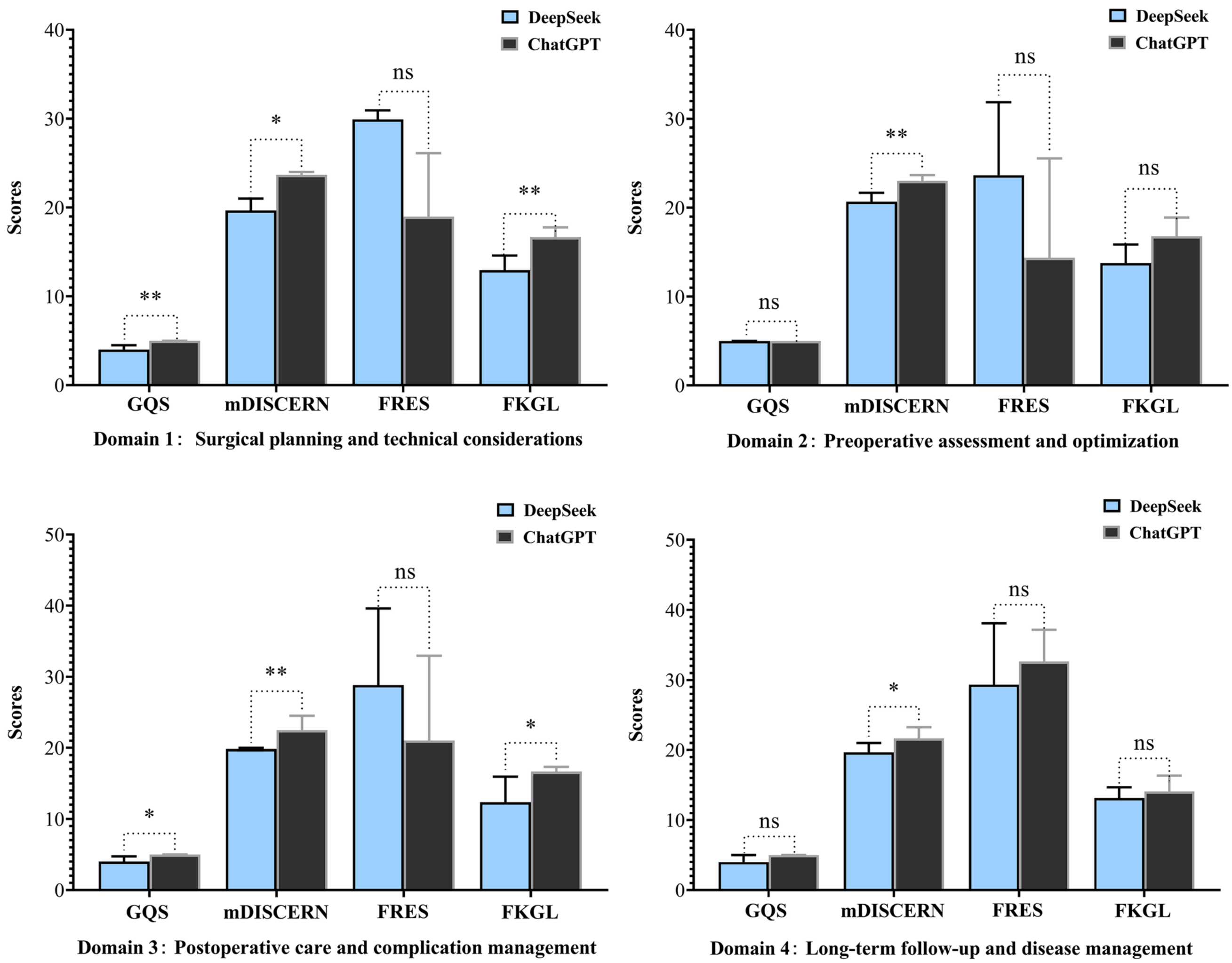

Fig. 4 Comparison of ChatGPT vs DeepSeek performance in each of four thematic domains (blue = DeepSeek, black = ChatGPT). Each subplot corresponds to one domain and displays the results for all four metrics: Global Quality Score (GQS), mDISCERN, Flesch Reading Ease Score (FRES), and Flesch-Kincaid Grade Level (FKGL). As the data included both normally and non-normally distributed variables, all values are expressed as medians with IQR; statistical significance is indicated as ns = not statistically, * = $P < 0.05$, ** = $P < 0.01$, *** = $P < 0.001$ (ChatGPT vs DeepSeek)

Discussion

Bariatric (metabolic) surgery is an effective treatment for severe obesity, achieving sustained weight loss and improvements in metabolic health and obesity-related conditions. With obesity rates and surgery numbers rising globally, advanced strategies are needed to optimize patient care and healthcare efficiency. AI-based interventions in bariatric patient education have gained increasing attention in recent years. First, AI can deliver personalized and accessible information, tailored to patients' literacy levels and cultural contexts, thereby supporting informed consent and shared decision-making. Second, AI tools can reinforce perioperative instructions and long-term lifestyle modifications, promoting adherence and outcomes. Third, by automating routine education and addressing frequently asked questions, AI can reduce clinician workload and enhance healthcare efficiency [16].

This study demonstrates that ChatGPT outperforms DeepSeek in providing factual, evidence-based information for bariatric patient education, consistently yielding more accurate and reliable responses as indicated by higher mDISCERN scores and qualitative content. These findings reflect ongoing advancements in AI capabilities, particularly in addressing structured, guideline-driven clinical queries [17]. Despite these improvements, both models generated content at a college-level reading standard, potentially limiting accessibility for patients with lower health literacy. Further optimization is needed to enhance the readability and inclusivity.

Despite their accuracy and organization, both models showed limited humanistic and emotionally supportive content. Responses prioritized clinical mechanisms, procedural details, and nutritional guidelines while largely omitting patient-centered issues such as emotional challenges, body image concerns, and the psychological impact of postoperative life [18]. For example, when asked, *"Is it normal to have a lot of loose skin after weight loss surgery?"*, neither model adequately acknowledged the associated emotional distress or provided empathetic guidance for psychosocial adaptation. Neither provided recommendations for mental health resources, peer support, or coping strategies. Although ChatGPT occasionally referenced multidisciplinary care and long-term commitment, these mentions lacked the depth to meaningfully support patients' lived experiences. This highlights a critical limitation in current AI-based patient education tools: the insufficient integration of humanistic and emotional considerations, which are essential components of holistic bariatric care [19]. Incorporating patient-centered design and empathetic dialogue frameworks may improve the suitability of future language models for real-world clinical support and education.

To ensure a rigorous assessment of AI-generated educational responses for bariatric patients, we utilized four widely validated metrics, each measuring distinct dimensions of text complexity. The GQS provided an efficient and intuitive tool for assessing the overall quality of AI-generated responses in bariatric education. Its simple 5-point structure enabled rapid differentiation between low- and high-quality content, particularly highlighting responses that were comprehensive, well-organized, and suitable for patient use [20, 21]. However, GQS has inherent limitations. Because it relies on subjective judgment, inter-rater reliability may be reduced, and it lacks explicit criteria for evaluating factual accuracy or source credibility [22]. As such, while GQS effectively captures presentation quality and user-friendliness, it is best complemented by more granular tools like DISCERN to ensure a holistic evaluation of medical information. The DISCERN tool proved valuable in evaluating the reliability and balance of AI-generated health information [23]. However, DISCERN is time-intensive and relies on expert interpretation, which limits its scalability in large-scale AI evaluations. Accordingly, the original 15-item DISCERN tool was streamlined into a 5-item version (mDISCERN) to enhance assessment efficiency and feasibility. Moreover, it does not address empathy, cultural adaptation, or emotional tone—factors increasingly essential in patient-centered communication [24]. The FKGL and FRES provide efficient, standardized measures of text readability, making them valuable for evaluating AI-generated health content. However, these tools assess form rather than meaning—overlooking medical jargon, logical flow, and cultural nuance [25]. While useful for screening, these indices should be paired with qualitative assessments to ensure true patient comprehension.

Regarding readability, both models generated content suited to college-level readers; however, neither adapted adequately for patients with lower health literacy. Simplifying language while preserving medical accuracy could significantly improve accessibility across broader patient populations [26]. This consideration is especially pertinent in the context of bariatric care, where the effective communication of complex clinical information to individuals of varying educational backgrounds is essential for informed decision-making and treatment adherence [27].

A key limitation of this study is the absence of direct patient input in evaluating the AI-generated responses. While assessments by a multidisciplinary panel of bariatric and metabolic surgeons provided a structured appraisal of response quality, reliability, and readability, they do not capture how patients interpret, understand, or use the content. Clinicians may prioritize medical completeness over accessibility, emotional resonance, or practical usefulness, so patient comprehension, perceived value, and satisfaction were not directly measured. To address this limitation and improve real-world applicability, future research should adopt mixed-methods frameworks that combine quantitative performance metrics with qualitative patient feedback [28]. Semi-structured interviews and focus groups with bariatric surgery patients can explore perceptions of clarity, emotional tone, and trustworthiness, while validated patient-reported outcome measures (PROMs)—such as the Patient Education Materials Assessment Tool (PEMAT) and the Suitability Assessment of Materials (SAM)—can assess comprehension, relevance, and satisfaction [29, 30]. Recruiting participants from postoperative clinics, bariatric support groups, and online patient communities will ensure diversity. This approach will refine AI-generated content and enhance its empathy, cultural sensitivity, and accessibility across different literacy levels.

In addition to the lack of patient-centered validation, the question set used in this study was derived exclusively from the ASMBS patient FAQs. While clinically relevant, this limited scope may not fully represent the range of patient concerns or expressions across broader or non-US populations. Similarly, the comparison was limited to two AI models—ChatGPT and DeepSeek—that differ in their language background, development context, and training data. These factors likely influenced model performance and limit the generalizability of our findings. Future studies should expand both the scope of patient questions and the range of AI models to better reflect diverse populations and healthcare settings.

To address these limitations, several key recommendations can be made for future development of AI applications in bariatric care. First, adaptive response systems should tailor information to patients' educational backgrounds and comprehension abilities. Second, culturally responsive communication frameworks should be integrated, reflecting variations in health literacy, language preferences, and cultural norms. Third, emotional intelligence components—capable of recognizing and responding to emotional cues—should be incorporated to enhance patient engagement [31]. Finally, multimodal delivery formats (e.g., text, audio, and visual aids) could further improve comprehension and accessibility across diverse populations [32].

A further limitation is the restricted transparency of the evaluated language models. ChatGPT and DeepSeek are both proprietary systems, and as such, key details regarding their architecture, parameter size, training corpora, and update histories are not publicly disclosed. This opacity limits reproducibility and interpretability across studies. Although we specified the model versions and access dates in the Methods section, future evaluations would benefit from including open-source or fully documented models to allow more rigorous validation and comparison.

Another limitation is the absence of systematic evaluation of potential bias, factual inaccuracy, or outdated content in AI-generated responses. Although the modified mDISCERN tool partially addressed issues related to source transparency and balance of information, it does not capture the spectrum of reliability concerns—particularly misinformation or omissions relative to clinical guidelines. Given that LLMs rely heavily on web-sourced and non-curated data, future research should include direct fact-checking of AI outputs against authoritative clinical guidelines to ensure accuracy and safety. Key references for bariatric surgery and obesity management include the ASMBS Clinical Practice Guidelines [33], the International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO) Global Guidelines [34], the National Institute for Health and Care Excellence (NICE) recommendations (UK) [35], and the National Institutes of Health (NIH) guidelines (US) [36]. These evidence-based frameworks provide standards for surgical indications, perioperative care, nutrition, and long-term follow-up, serving as benchmarks for evaluating AI-generated content. Structured coding schemes can further identify inaccuracies, omissions, or outdated information.

Beyond accuracy, ethical safeguards are essential for deploying AI in patient education. These include strict data privacy protections, informed consent when patient data or interactions are involved, and transparency about the non-human origin of responses [37, 38]. Clear accountability frameworks are also needed to assign responsibility for errors or misinformation. Such measures are critical to maintain patient trust, particularly in sensitive fields like bariatric surgery, where emotional vulnerability and decision-making complexity are high [39].

Beyond the scope of bariatric surgery, future studies should assess LLM performance in broader patient education domains, particularly in real-time, interactive clinical settings. Advancements should aim to enhance contextual sensitivity and psychosocial responsiveness [40]. Embedding both factual accuracy and emotional awareness into AI-generated communication may help these tools function as more effective adjuncts to clinicians, supporting comprehensive, patient-centered care.

Conclusion

This study found that ChatGPT generally produced higher-quality and more reliable responses for bariatric surgery education than DeepSeek, although DeepSeek's simpler language did not translate into better readability. Both models lacked emotional sensitivity and patient-centered nuance, especially in addressing psychosocial and cultural aspects, underscoring challenges in replicating holistic clinical communication.

AI-generated responses may provide structured medical information and serve as a supplementary resource, but current limitations—college-level readability and minimal psychosocial engagement—restrict their applicability for patients with diverse literacy levels. Greater simplification, personalization, and integration of empathy and culturally responsive communication will be essential for future tools to support inclusive and effective digital health education.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11695-025-08249-x.

Author Contribution S.G., C.-L. Y. and Q.W. contributed to the idea and creation of the article, S.G., X.-P. L. and M.J. wrote the main manuscript text, J.C., K.-X.T. and W.-W.Y. prepared all figures and tables, X.-R. J. and P.L. collected and statistically analyzed the data, all authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing interests The authors declare no competing interests.

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